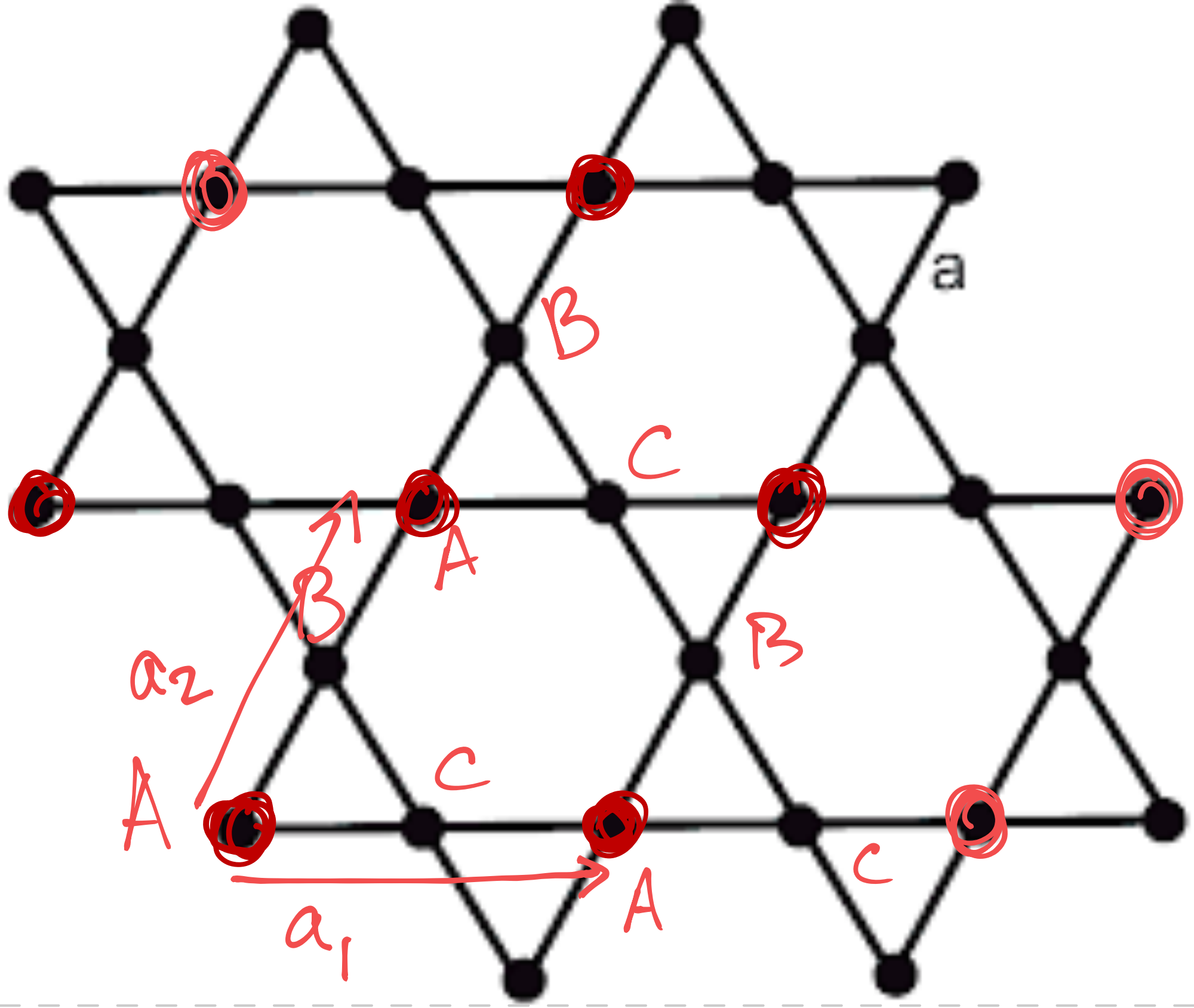

a
B
C
A
B
B
a_2
A
C
C
A
a_1

3 atom per unit cell. Unit cell = up-pointing triangle.

Primitive lattice vectors $\vec{a}_1 = (1,0)\,2a$; $\vec{a}_2 = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) 2a$.

Area of the unit cell = area of the minimal rombus formed by A sites:

$$\text{area} = |\vec{a}_1 \times \vec{a}_2| = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 1 & 0 & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \end{vmatrix} (2a)^2 = (2a)^2 \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}(2a)^2.$$

Reciprocal lattice $\vec{b}_1 = 2\pi \dfrac{\vec{a}_2 \times \vec{a}_3}{\vec{a}_1 \cdot \vec{a}_2 \times \vec{a}_3}$

$$\vec{b}_2 = 2\pi \frac{\vec{a}_3 \times \vec{a}_1}{\vec{a}_1 \cdot \vec{a}_2 \times \vec{a}_3}$$

Need to add $\vec{a}_3 = (0,0,1)\,a$.

$$\vec{a}_1 \cdot \vec{a}_2 \times \vec{a}_3 = \begin{vmatrix} 2 & 0 & 0 \\ 1 & \sqrt{3} & 0 \\ 0 & 0 & 1 \end{vmatrix} a^3 = 2\sqrt{3}\, a^3.$$

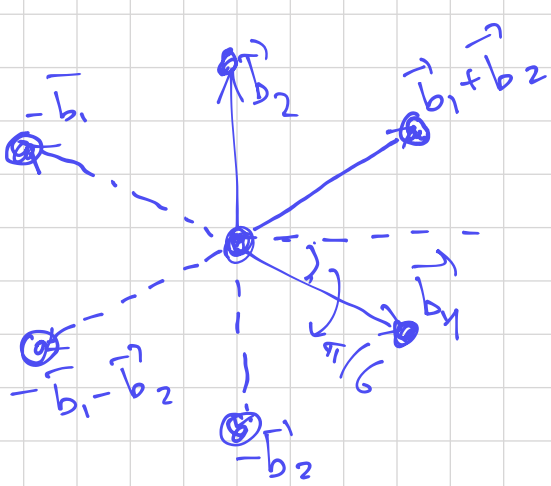
$$\vec{a}_2 \times \vec{a}_3 = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 1 & \sqrt{3} & 0 \\ 0 & 0 & 1 \end{vmatrix} a^2 = a^2 (\hat{x}\sqrt{3} - \hat{y})$$

$$\vec{a}_3 \times \vec{a}_1 = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0 & 0 & 1 \\ 2 & 0 & 0 \end{vmatrix} a^2 = a^2 (-\hat{y})(-2) = 2a^2 \hat{y}.$$

$$\Rightarrow \begin{cases} \vec{b}_1 = 2\pi \dfrac{(\sqrt{3}\hat{x} - \hat{y})a^2}{2\sqrt{3}a^3} = \dfrac{2\pi}{a}\left(\dfrac{1}{2}\hat{x} - \dfrac{1}{2\sqrt{3}}\hat{y}\right) = \dfrac{2\pi}{(2a)}\left(\hat{x} - \dfrac{1}{\sqrt{3}}\hat{y}\right) \\ \vec{b}_2 = 2\pi \dfrac{2a^2\hat{y}}{2\sqrt{3}a^3} = \dfrac{2\pi}{a}\dfrac{1}{\sqrt{3}}\hat{y} = \dfrac{2\pi}{(2a)}\dfrac{2}{\sqrt{3}}\hat{y} \end{cases}$$

We are using units of $\frac{2\pi}{(2a)}$ because $|\vec{a}_1| = |\vec{a}_2| = 2a$.

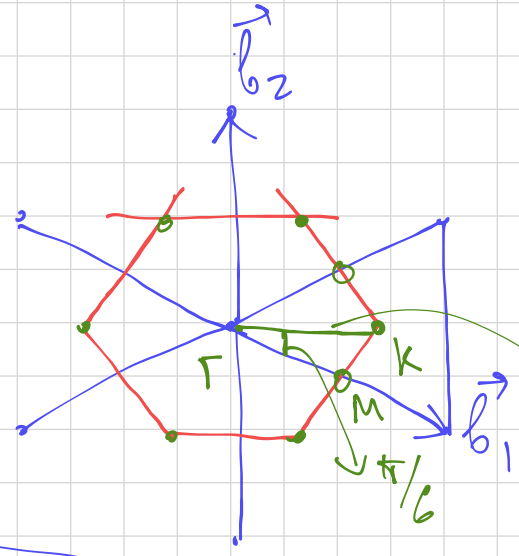
Therefore $\vec{b}_{1,2}$ are forming triangular lattice (Prob. 13.6(b)).



$$\vec{b}_1 = \frac{2\pi}{2a}\frac{2}{\sqrt{3}}\left(\frac{\sqrt{3}}{2}\hat{x} - \frac{1}{2}\hat{y}\right)$$

$\Rightarrow$ vector makes $\pi/6$ with $\hat{x}$ axis

Every site of the reciprocal lattice is surrounded by 6 other sites $\Rightarrow$ it is <u>triangular</u> lattice.



Triangular reciprocal lattice.

Red hexagon = 1 BZ

$$\underbrace{\Gamma K \cos\frac{\pi}{6}}_{\frac{\sqrt{3}}{2}} = \underbrace{\frac{1}{2}\left(\frac{2\pi}{2a}\,\frac{2}{\sqrt{3}}\right)}_{\Gamma M} \Rightarrow \Gamma K = \frac{1}{2}\,\frac{2\pi}{2a}\,\frac{4}{3} = \frac{2\pi}{2a}\,\frac{2}{3}$$

$$\text{Area } \Gamma K M = \frac{1}{2}(\Gamma K)\cdot(\Gamma M)\cdot \sin\frac{\pi}{6}$$

$$= \frac{1}{2}\left(\frac{2\pi\cdot 2}{2a\cdot 3}\right)\left(\frac{\pi}{2a}\,\frac{2}{\sqrt{3}}\right)\cdot\frac{1}{2} = \left(\frac{2\pi}{2a}\right)^2\frac{1}{6\sqrt{3}}$$

Check:

$$\text{area BZ} = |\vec{b}_1\times\vec{b}_2| = \left(\frac{2\pi}{2a}\right)^2\frac{2}{\sqrt{3}}$$

$$\text{Area BZ} = 12\,(\Gamma K M) = \frac{12}{6\sqrt{3}}\left(\frac{2\pi}{2a}\right)^2 = \frac{2}{\sqrt{3}}\left(\frac{2\pi}{2a}\right)^2$$

So that $(\text{Area BZ})\cdot(\text{Area unit cell}) = \frac{2}{\sqrt{3}}\left(\frac{2\pi}{2a}\right)^2\cdot\frac{\sqrt{3}}{2}(2a)^2 = (2\pi)^2$

as it should be.

There are 3 bands so that for each $\vec{k}\in 1BZ$, one has 3 possible energy bands $\varepsilon_1(k), \varepsilon_2(k), \varepsilon_3(k)$.

Assuming the bands do not overlap (they do touch, but only at a few special $\vec{k}$-points) we can order them as

$$\varepsilon_1 < \varepsilon_2 < \varepsilon_3.$$

Total number of states inside $1BZ = \underbrace{2}_{\text{spin}} \times \underbrace{N_A}_{\text{number of unit cells = number of sites A}} \times \underbrace{3}_{\text{number of bands}}$

Since each of A, B, C sites gives 1 electron, there is a total of $3N_A$ electrons.

Therefore band $\varepsilon_1(\vec{k})$ is fully filled, band $\varepsilon_2(\vec{k})$ is ½-filled and band $\varepsilon_3(\vec{k})$ is empty. ⇒ chemical potential is crossing ε_2 band